Interpreting Personality Prediction: Word-Level vs Sentence-Level Contributions in Neural Models

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Abstract

This project studies interpretability reliability in neural personality prediction on MBTI-500 text data. We compare attention-, gradient-, and perturbation-based explanations at word and sentence levels to test agreement across methods, local versus global encoding behavior, and sensitivity to spurious topic/style cues. The outcome is a reproducible evaluation plan that moves beyond classification accuracy and tests whether explanation claims are stable and meaningful.

Keywords: personality prediction, interpretability, MBTI, NLP, deep learning

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1 Introduction

Accurate personality prediction from text can support adaptive learning systems, mental-health triage support, personalized recommendation, and human-computer interaction design. However, in high-stakes settings, predictive performance alone is not enough; we also need interpretable evidence for why a model predicts a personality label. Our project is motivated by this gap between model accuracy and explanation quality for personality inference from user-generated text.

Existing personality prediction systems largely optimize classification metrics while offering limited analysis of what textual evidence drives predictions. Even when interpretability methods are applied, studies often report one explanation method in isolation and mostly at a single granularity (token-level or feature-level), making it difficult to assess reliability. Prior work also does not sufficiently test whether models use meaningful personality cues versus spurious correlations such as topic words or social-media style markers.

We propose a comparative interpretability study on the MBTI-500 dataset [6] using transformer-based neural models [1]. Our central research question is: *How do neural models attribute importance to words versus sentences when predicting personality, and are these attributions consistent across explanation methods?* We will compare attention-, gradient-, and perturbation-based explanations, test local (word) versus global (sentence) encoding hypotheses, and evaluate robustness to spurious cues. This is new in combining cross-method agreement with multi-granularity ablations and spurious-signal tests in one unified experimental pipeline.

2 Related Work

2.1 Topic 1: Neural Personality Prediction from Text

Majumder et al. [8] introduce deep document modeling for personality detection and show neural models can outperform traditional feature-engineering baselines on text-based personality tasks. Gjurković and Šnajder [3] demonstrate that Reddit language provides useful behavioral signals for personality prediction and highlight the value of social-media corpora for this problem. Majumder et al. [9] further emphasize end-to-end deep learning for personality inference, reinforcing that learned representations improve predictive quality over shallow approaches.

Our work is different because it focuses on *interpretability reliability* (agreement, granularity sensitivity, and spurious-cue dependence) rather than only predictive performance.

2.2 Topic 2: Feature Attribution and Explanation Methods in NLP

Ribeiro et al. [10] propose LIME, a perturbation-based local explanation method that approximates model behavior around an instance to identify influential features. Sundararajan et al. [13] introduce Integrated Gradients, a gradient-based attribution method grounded in axioms such as sensitivity and implementation invariance. Lundberg and Lee [7] provide SHAP, a unified attribution framework based on Shapley values with strong consistency guarantees and broad model applicability.

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Our work is different because we apply these explanation families jointly to personality prediction and explicitly quantify *cross-method consistency* for both words and sentences.

2.3 Topic 3: Reliability of Attention and Faithfulness of Explanations

Jain and Wallace [5] show that attention distributions are often weakly correlated with other feature-importance measures and can be manipulated without major prediction changes. Serrano and Smith [12] test attention faithfulness via erasure analyses and report that high attention weights do not always correspond to causally important tokens, challenging the assumption that attention is intrinsically explanatory.

Our work is different because we do not evaluate one explanation channel in isolation; we directly test where methods agree or disagree and connect disagreement to downstream behavior via removal experiments.

2.4 Topic 4: Spurious Correlations and Shortcut Learning

Gururangan et al. [4] identify annotation artifacts that allow models to exploit superficial cues, demonstrating how dataset biases can inflate apparent benchmark performance. Geirhos et al. [2] formalize shortcut learning and show that deep models often rely on easy but non-robust patterns instead of task-relevant semantics. Sagawa et al. [11] analyze spurious correlations under overparameterization and show that standard empirical risk minimization can fail on minority or distribution-shifted groups.

Our work is different because we operationalize shortcut analysis for personality prediction by removing topic-heavy and stylistic markers, then measuring prediction shifts and performance changes.

3 Methods

3.1 Problem Setup and Data

We use the MBTI-500 dataset [6], where each instance contains text and one of 16 MBTI labels. We frame this as a 16-class classification task and also report binary splits per trait dimension (E/I, S/N, T/F, J/P) for additional analysis.

3.2 Model and Training Configuration

We fine-tune a BERT-base encoder [1] with a classification head. Text is segmented into sentences, tokenized with WordPiece, and truncated/padded to a fixed maximum length. We perform train/validation/test splits with fixed random seeds, use early stopping on validation macro-F1, and report mean and standard deviation across 3 runs. Hyperparameters (learning rate, batch size, max length, epochs) are selected by validation performance.

3.3 Explanation Pipeline

For each test sample, we compute:

- **Attention-based importance:** aggregated attention scores for tokens and sentence-level pooled scores.
- **Gradient-based importance:** token-level integrated gradients [13], then sentence scores by averaging token attributions within each sentence.
- **Perturbation-based importance:** prediction change after masking top candidate words or sentences.

Importance scores are normalized per sample, then ranked to obtain top-k words and top-k sentences. This enables method-to-method comparison at both granularity levels.

3.4 Research Questions

- **RQ1:** Do different interpretability methods (attention, gradients, perturbation) agree on what parts of text are important for personality prediction?
- **RQ2:** Is personality encoded in localized signals (specific words) or distributed across higher-level structures (sentences)?
- **RQ3:** Are models relying on genuine personality cues or spurious correlations such as topics or slang?
- **RQ4:** Do human judgments of important words/sentences align with model explanations?

4 Experiments

4.1 Experiment 1: Cross-Method Agreement Analysis

Main purpose: Quantify whether attention, gradient, and perturbation explanations are consistent about which words/sentences matter for personality prediction. This validates whether interpretability conclusions are method-dependent. This experiment directly evaluates **RQ1** by measuring agreement across interpretability methods.

Procedure:

- For each method and sample, extract top-k words and top-k sentences.
- Compute pairwise agreement between methods at word and sentence levels.
- Aggregate by class and overall dataset.

Evaluation metrics:

- **Jaccard similarity** on top-k sets (overlap of highlighted evidence).
- **Spearman correlation** on full ranked importance scores (global ranking agreement).

Jaccard captures discrete overlap, while Spearman captures monotonic ranking consistency.

4.2 Experiment 2: Granularity, Spurious-Cue, and Human-Alignment Analysis

Main purpose: Test whether personality evidence is local or distributed, whether model behavior depends on non-causal

shortcuts (topic/slang cues), and whether model-identified important text aligns with human judgment.

Procedure:

- Remove top-k important words and re-evaluate.
- Remove top-k important sentences and re-evaluate.
- Remove predefined topic-heavy words and stylistic markers, then re-evaluate.
- Compare with random-removal baselines matched for deletion size.
- For a sampled subset, collect human-annotated important words/sentences and compare these with model explanation outputs.

Evaluation metrics:

- **Macro-F1 and accuracy drop** after each removal setting (tests information concentration).
- **Prediction flip rate** after spurious-cue removal (tests shortcut dependence).
- **Relative degradation gap** between sentence-removal and word-removal (tests local vs global encoding).
- **Human-model agreement** using top-k overlap percentage and Jaccard similarity between human-selected and model-selected words/sentences (tests explanation alignment with human judgment).

These metrics directly map to RQ2, RQ3, and RQ4 and are reproducible via fixed seeds and fixed deletion budgets.

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